Likelihood-Based Inference in Counting Experiments

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1 Purpose of this note

This note introduces likelihood-based parameter estimation, with a focus on counting experiments. The goal is to clarify how statistical assumptions enter a fit, how they affect parameter estimates and uncertainties, and why likelihood methods are often preferable to simple curve fitting.

A central theme throughout is:

> Statistical inference requires *both* a model for the underlying signal and a model for how data fluctuate around that signal.

We typically only think about the first part, writing down a peak shape or a background model. The second part, the statistical model, is where things tend to get hand-wavy.

I also want to point out that while the concept of likelihood is both frequentist and Bayesian, the interpretation I use here from the frequentist point of view.

2 TL;DR

This not is very long, and if you don't read it all, you should feel bad about yourself. However, the punchline is this: **if you don't understand your model and/or your noise and use the incorrect description of your data, your uncertainty estimates will be WRONG and will often UNDERESTIMATE your error.** As a second point, 'goodness-of-fit' metrics are often abused, misused, and misunderstood.

If you would like to understand why this is the case (and you *should* want to), please read on.

3 From probability to likelihood

Suppose we observe data $\{y_i\}$, and we wish to infer parameters $\boldsymbol{\theta}$ describing an underlying model.

The starting point is a **probability model**:

$$P(y_i \mid \boldsymbol{\theta}), \tag{1}$$

which specifies how likely it is to observe data y_i given parameters $\boldsymbol{\theta}$.

Assuming independent measurements, the joint probability is:

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_i P(y_i \mid \boldsymbol{\theta}). \tag{2}$$

We interpret this as a function of $\boldsymbol{\theta}$ (with the data fixed), and call it the **likelihood**.

The principle of maximum likelihood states that the best parameter estimates are:

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}). \tag{3}$$

This is referred to as Maximum Likelihood Estimations, or **MLE**.

In practice, we maximize the log-likelihood (or equivalently minimize the negative log-likelihood):

$$\log \mathcal{L}(\boldsymbol{\theta}) = \sum_i \log P(y_i \mid \boldsymbol{\theta}). \tag{4}$$

3.1 The generative modeling viewpoint

A useful way to think about this is through a **generative model**. The idea is that the data were produced by a two-step process:

1. The model predicts a quantity $\mu_i(\boldsymbol{\theta})$.
2. The data are generated as random fluctuations around μ_i.

So the full model has two ingredients:

- a deterministic prediction $\mu_i(\boldsymbol{\theta})$, e.g. the "fit function,"
- a statistical rule describing fluctuations around μ_i.

This distinction is important, as function $\mu_i(\boldsymbol{\theta})$ is not just a convenient curve to draw through data points:

$$\boxed{\mu_i(\boldsymbol{\theta}) = \text{expected value of the measurement in bin } i.} \tag{5}$$

The likelihood then encodes how measurements fluctuate around this expectation. The choice of likelihood (Gaussian, Poisson, etc) is a statement about the measurement process, not just a mathematical convenience.

4 Gaussian likelihood and least squares

Before getting to counting experiments, it is worth seeing where the familiar least-squares approach comes from, since this is what most of us use by default.

4.1 The model

Suppose each measurement y_i is a continuous quantity with Gaussian noise:

$$y_i = f(x_i; \boldsymbol{\theta}) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_i^2). \tag{6}$$

The probability of observing y_i is:

$$P(y_i \mid \boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{(y_i - f(x_i; \boldsymbol{\theta}))^2}{2\sigma_i^2}\right]. \tag{7}$$

The log-likelihood for independent measurements is:

$$\log \mathcal{L} = -\frac{1}{2} \sum_i \left[\frac{(y_i - f(x_i; \boldsymbol{\theta}))^2}{\sigma_i^2} + \log(2\pi\sigma_i^2)\right]. \tag{8}$$

Since the second term does not depend on $\boldsymbol{\theta}$, maximizing the likelihood is equivalent to minimizing:

$$\chi^2 = \sum_i \frac{(y_i - f(x_i; \boldsymbol{\theta}))^2}{\sigma_i^2}. \tag{9}$$

$$\boxed{\text{Weighted least squares is maximum likelihood under the assumption of Gaussian noise with known variances.}} \tag{10}$$

Each data point is weighted by $w_i = 1/\sigma_i^2$: points with smaller uncertainty pull harder on the fit, and points with larger uncertainty contribute less.

If all uncertainties are equal ($\sigma_i = \sigma$), this reduces to unweighted least squares: $\chi^2 \propto \sum_i (y_i - f_i)^2$.

4.2 The key assumption

This framework assumes that the variances σ_i^2 are known and independent of the model prediction. For many types of data, this is a reasonable assumption.

But in counting experiments, the variance depends on the expected signal:

$$\mathrm{Var}(n_i) = \mu_i. \tag{11}$$

This violates the assumptions underlying weighted least squares. The variance is not a known constant—it is part of what we are trying to estimate. This is the fundamental reason we need a different approach for count data.

5 Counting experiments and Poisson statistics

In many experiments, the data consist of counts n_i in bins. Examples include photon counts in detectors, particle counts in high-energy physics, and diffraction intensities. These are the cases where the distinction between curve fitting and likelihood fitting becomes especially important.

A standard model for count data is:

$$n_i \sim \mathrm{Poisson}(\mu_i), \tag{12}$$

where μ_i is the expected number of counts in bin i.

The Poisson distribution has:

$$\mathbb{E}[n_i] = \mu_i, \quad \mathrm{Var}(n_i) = \mu_i. \tag{13}$$

This tells us two things:

- bins with larger counts fluctuate more,
- the variance is not constant across the dataset.

5.1 The Poisson likelihood

The probability of observing n_i counts when μ_i are expected is:

$$P(n_i \mid \mu_i) = \frac{\mu_i^{n_i} e^{-\mu_i}}{n_i!}. \tag{14}$$

The total log-likelihood across all bins is:

$$\log \mathcal{L} = \sum_i \left[n_i \log \mu_i - \mu_i - \log(n_i!) \right]. \tag{15}$$

The $\log(n_i!)$ terms depend only on the data, not on the parameters $\boldsymbol{\theta}$, so they do not affect the optimization. Dropping them, we minimize:

$$-\log \mathcal{L} = \sum_i \left[\mu_i - n_i \log \mu_i \right]. \tag{16}$$

$$\boxed{\text{The model prediction } \mu_i(\boldsymbol{\theta}) \text{ is the Poisson mean.}} \tag{17}$$

The fit function is therefore not separate from the statistical model. It supplies the mean value around which the observations fluctuate.

5.2 When can you get away with Gaussian instead?

For large expected counts ($\mu_i \gtrsim$ 20–30), the Poisson distribution is well approximated by a Gaussian with $\sigma_i = \sqrt{\mu_i}$. In this regime, the Poisson likelihood and weighted χ^2 with $\sigma_i = \sqrt{\mu_i}$ give nearly identical results.

The difference matters for low-count bins. If your spectrum has bins with fewer than ~ 10 counts, or bins with zero counts, you should use the Poisson likelihood. The Gaussian approximation handles these bins poorly since it assigns zero weight to empty bins (since $\sigma = \sqrt{0} = 0$, which is undefined) and treats all low-count bins as if their uncertainties were symmetric, *which they are not.*

6 Why least squares can fail for count data

It is common practice to fit count data using weighted least squares with $\sigma_i = \sqrt{n_i}$ (the observed counts):

$$\chi^2 = \sum_i \frac{(n_i - \mu_i)^2}{n_i}. \tag{18}$$

This is often called "Neyman's χ^2." An alternative uses the model prediction in the denominator:

$$\chi^2 = \sum_i \frac{(n_i - \mu_i)^2}{\mu_i}, \tag{19}$$

which is "Pearson's χ^2."

Both are reasonable approximations when counts are large. But both have problems when counts are small:

- Neyman's χ^2: bins with $n_i = 0$ contribute $\mu_i^2/0$, which is undefined. In practice these bins are dropped or assigned an arbitrary weight, *which biases the fit.*
- Pearson's χ^2: somewhat better, but the variance estimate $\sigma_i^2 = \mu_i$ depends on the quantity being estimated, so the weights change during the fit. This can cause convergence issues and subtle biases. (note, from much first hand experience, this is a pain in the ass....)
- Both least-squares approaches assume symmetric fluctuations about the mean. This is not accurate for Poisson data at low counts. The Poisson distribution is bounded below by zero and becomes strongly skewed when μ is small, so upward and downward fluctuations of the same magnitude do not have equal probability.

The Poisson likelihood handles all of these cases correctly, because it is the actual probability model for the data. It does not need ad hoc fixes for empty bins or asymmetric fluctuations because these are built into the Poisson distribution itself.

7 Practical fitting

In practice, you hand '$-\log \mathcal{L}(\boldsymbol{\theta})$' to a numerical optimizer, e.g. `scipy.optimize.minimize` or `scipy.optimize.curve_fit` (The latter is specifically designed for least-squares problems but does not directly support Poisson likelihoods.)

For Poisson fits, the typical workflow is:

1. Write a function that computes $\mu_i(\boldsymbol{\theta})$ for a given set of parameters.
2. Write a function that computes $-\log \mathcal{L} = \sum_i [\mu_i - n_i \log \mu_i]$.
3. Pass this to `scipy.optimize.minimize` with initial parameter guesses.
4. The optimizer returns $\hat{\boldsymbol{\theta}}$ (the best-fit parameters).

Libraries like `iminuit` (a Python interface to the MINUIT minimizer widely used in particle physics) make this particularly convenient and also provide built-in tools for uncertainty estimation via profile likelihood.

8 Application: peak fitting in XRD

To make things concrete, consider X-ray diffraction (XRD), where the data are photon counts as a function of scattering angle 2θ.

Since this is count data, we use the Poisson likelihood. The question is: what is the model $\mu_i(\boldsymbol{\theta})$?

A reasonable model is:

$$\mu(x; \boldsymbol{\theta}) = P(x; \boldsymbol{\theta}_{\text{peak}}) + B(x; \boldsymbol{\theta}_{\text{bkg}}), \tag{20}$$

where $P(x)$ is a peak shape (e.g. a Voigt profile) and $B(x)$ is a background model (e.g. a flat background or low-order polynomial). Then:

$$\mu_i = \mu(x_i; \boldsymbol{\theta}) \tag{21}$$

is the expected count in bin i, and $n_i \sim \text{Poisson}(\mu_i)$.

The plotted fit curve is not merely descriptive, it is the statistical expectation entering the likelihood. Every point on that curve is a statement about the expected number of photons at that angle, given the model.

8.1 Comparison: Poisson likelihood vs least squares

Figure 1 shows both methods applied to the same XRD peak. Panel (a) shows the data with both fits overlaid. On a log scale, the difference is most visible in the tails, where counts are low and the Poisson likelihood treats empty and low-count bins according to their actual counting probabilities.

Panel (b) compares the profile likelihood curves for the peak position x_0. Both methods find similar best-fit values, but the shape of the profile (and corresponding uncertainty estimate) differs. The Poisson profile is the statistically correct one for count data.

Panels (c) and (d) show the standardized residuals. Under the Poisson model, the residuals $(n_i - \mu_i)/\sqrt{\mu_i}$ should scatter uniformly with unit variance if the model is adequate. Also, notice the 'bands', which is exactly a consequence of the very low bin count on the tails.

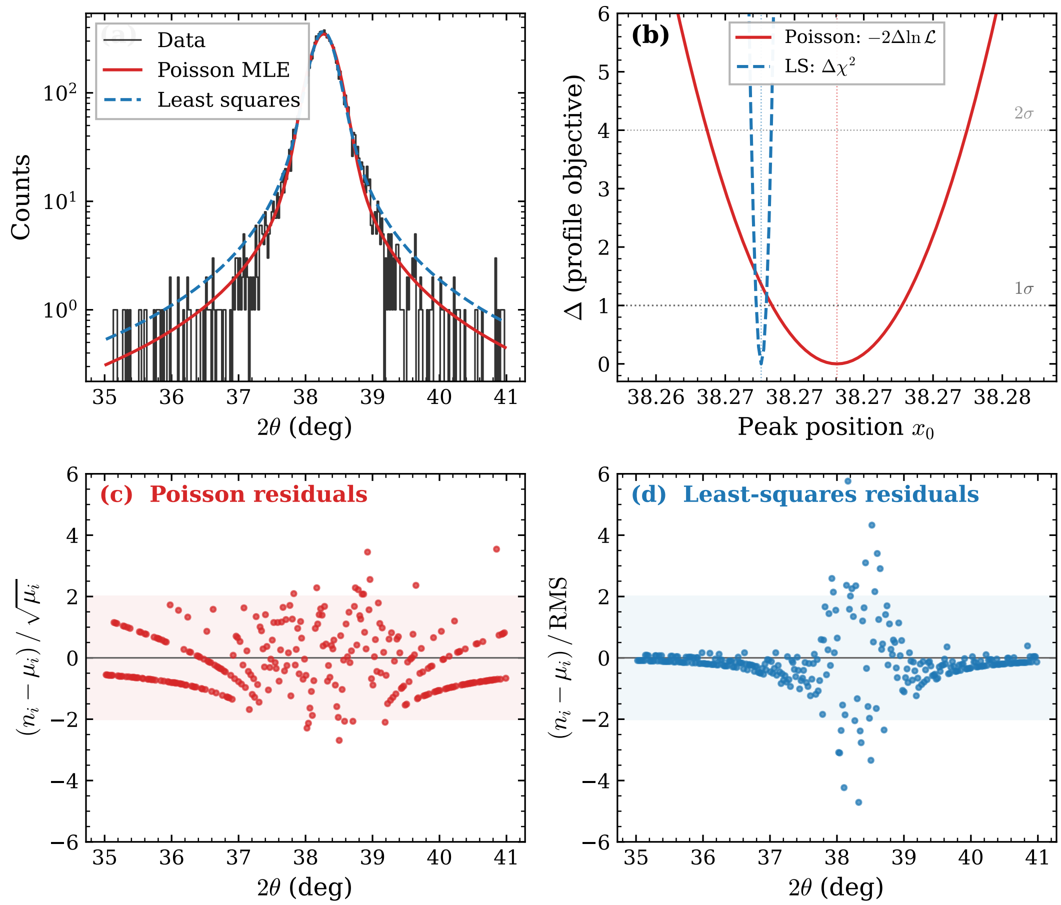

Figure 1: Comparison of Poisson likelihood and unweighted least-squares fits to an XRD peak. (a) Data and best-fit models. (b) Profile likelihood for the peak position x_0; horizontal lines mark the 1σ and 2σ thresholds. **Notice how the least squares estimate underestimates the uncertainty!!!** (c) Poisson-standardized residuals. (d) Least-squares residuals normalized by the global RMS.

9 Parameter uncertainties from the covariance matrix

After finding the best-fit parameters, we need uncertainties. The simplest approach is based on the local curvature of the log-likelihood near the optimum.

9.1 Quadratic approximation

Near the maximum, we expand the log-likelihood as a Taylor series:

$$\log \mathcal{L}(\boldsymbol{\theta}) \approx \log \mathcal{L}(\hat{\boldsymbol{\theta}}) - \frac{1}{2}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T H(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}), \tag{22}$$

where H is the **Hessian**:

$$H_{ij} = -\left.\frac{\partial^2 \log \mathcal{L}}{\partial\theta_i \partial\theta_j}\right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}. \tag{23}$$

This says that near the best-fit point, the log-likelihood looks like a multidimensional parabola. Equivalently, the likelihood itself is approximately a multivariate Gaussian in parameter space. This approximation for an example loss function can be seen in Fig. 2.

9.2 The covariance matrix

If the likelihood is approximately Gaussian, then the parameter estimates fluctuate (across hypothetical repeated experiments) with covariance:

$$\boxed{\mathrm{Cov}(\boldsymbol{\theta}) \approx H^{-1}.} \tag{24}$$

The diagonal elements give variances ($\sigma^2_{\theta_i} = (H^{-1})_{ii}$) and the off-diagonal elements encode correlations between parameters.

When a fitting routine reports a covariance matrix, it is usually based on this curvature approximation, either computed internally or estimated numerically near the optimum.

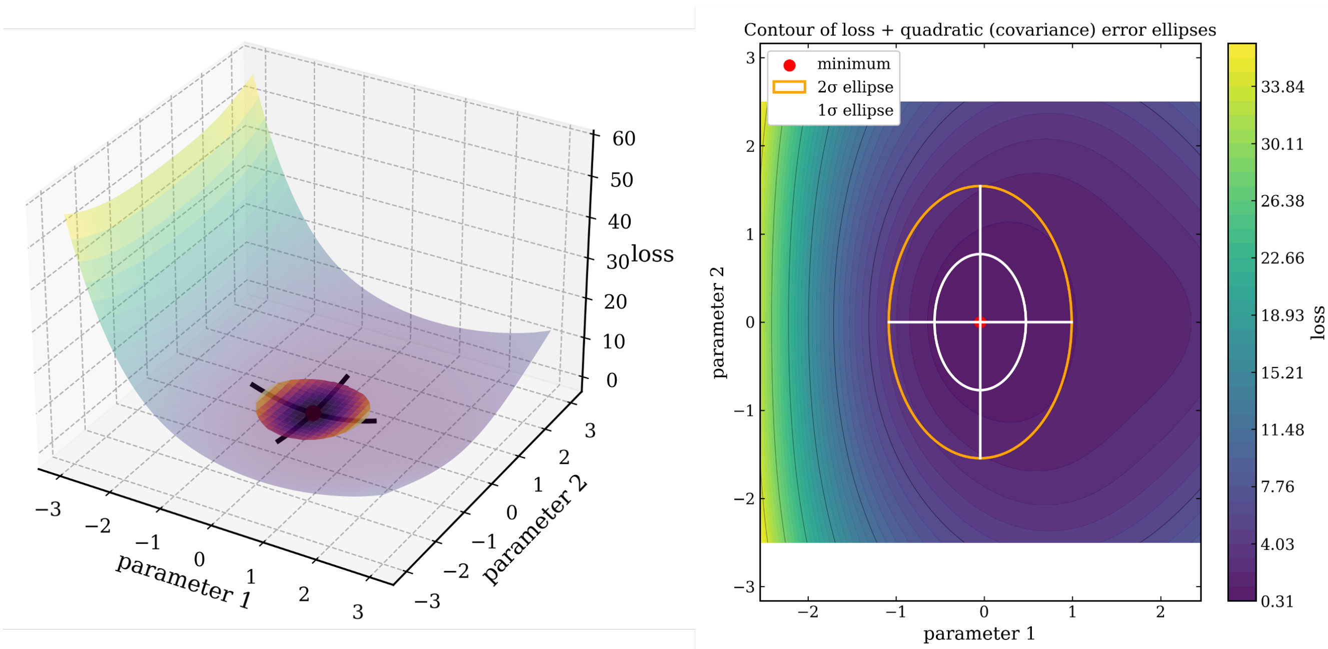

Figure 2: Example of a quadratic approximation of a two dimensional loss function surface.

9.3 The $\Delta = 1$ rule

An equivalent way to state this: if we define

$$\Delta(\boldsymbol{\theta}) = -2\left[\log \mathcal{L}(\boldsymbol{\theta}) - \log \mathcal{L}_{\max}\right], \tag{25}$$

then under the quadratic approximation, the 1σ region in parameter space corresponds to $\Delta = 1$. The contours of constant Δ are ellipsoids, and the principal axes of these ellipsoids define the parameter uncertainties and correlations.

9.4 What correlations mean

Non-zero off-diagonal elements in the covariance matrix mean that a change in one parameter can be partially compensated by a change in another without significantly degrading the fit.

In peak fitting, common correlations include:

- amplitude vs. width (a broader, shorter peak can look similar to a narrower, taller one),
- peak position vs. background (shifting the peak can be partially absorbed by the background),

- peak separation vs. amplitude ratio (in multi-peak fits).

These correlations are real features of the problem space and they reflect genuine ambiguities in what the data can tell you.

9.5 Limitations

The covariance matrix is a *local* approximation. It works well when the log-likelihood is close to parabolic near the minimum. It can fail when:

- the likelihood is asymmetric (common at low statistics),
- parameters are near physical boundaries (e.g. amplitudes that must be positive),
- there are strong degeneracies between parameters.

When the covariance matrix is not adequate, we need the profile likelihood.

10 Profile likelihood

10.1 The problem

In most fits, the full model depends on many parameters:

$$\boldsymbol{\theta} = (\psi, \boldsymbol{\eta}), \tag{26}$$

where ψ is the **parameter of interest** (e.g. peak position) and $\boldsymbol{\eta}$ are **nuisance parameters** (e.g. amplitude, width, background).

We want a reliable uncertainty on ψ. A common but incorrect approach is to fix all nuisance parameters at their best-fit values and vary ψ alone. This underestimates the uncertainty, because in reality, when ψ changes, the nuisance parameters are free to adjust to maintain a good fit.

10.2 Definition

The profile likelihood fixes ψ at a trial value and re-optimizes all nuisance parameters:

$$\boxed{\mathcal{L}_{\text{prof}}(\psi) = \max_{\boldsymbol{\eta}} \mathcal{L}(\psi, \boldsymbol{\eta}).} \tag{27}$$

This produces a one-dimensional function of ψ that accounts for all correlations with the nuisance parameters.

The procedure is: for each value of ψ on a grid, run a full fit of all other parameters, and record the best likelihood achieved. The result traces the *ridge* of the likelihood surface as ψ varies.

10.3 Interpretation

The profile likelihood answers the question:

> If the true value of ψ were this, how well could the model explain the data, after allowing all other parameters to adjust optimally?

This is fundamentally different from holding other parameters fixed. Fixing nuisance parameters artificially constrains the fit, making the likelihood drop faster as ψ moves away from the optimum, which makes the parameter look more precisely determined than it actually is.

10.4 Confidence intervals

Define the profile likelihood ratio:

$$\Delta(\psi) = -2\left[\log \mathcal{L}_{\text{prof}}(\psi) - \log \mathcal{L}_{\text{max}}\right]. \tag{28}$$

Under general conditions (Wilks' theorem), this quantity is approximately χ^2-distributed with 1 degree of freedom. The confidence interval is obtained by solving:

$$\Delta(\psi) = 1 \quad (\text{for } 1\sigma), \qquad \Delta(\psi) = 4 \quad (\text{for } 2\sigma), \qquad \Delta(\psi) = 9 \quad (\text{for } 3\sigma). \tag{29}$$

10.5 Relation to the covariance matrix

When the likelihood is exactly quadratic, the profile likelihood reduces to:

$$\Delta(\psi) = \frac{(\psi - \hat{\psi})^2}{\sigma_\psi^2}, \tag{30}$$

where $\sigma_\psi^2 = (H^{-1})_{\psi\psi}$. So in the Gaussian limit, the profile likelihood and the covariance matrix give the same answer.

$$\boxed{\text{The covariance matrix is a local approximation to the profile likelihood.}} \tag{31}$$

The profile likelihood is therefore a less restrictive uncertainty estimate. It does not assume the likelihood is symmetric, quadratic, or weakly correlated. It directly evaluates how the fit quality degrades as ψ moves away from the optimum.

10.6 Asymmetric uncertainties

Because the profile likelihood does not assume symmetry, it naturally produces asymmetric error bars:

$$\psi = \hat{\psi}^{+\sigma_+}_{-\sigma_-}. \tag{32}$$

This is common when parameters are near boundaries, when the model is nonlinear, or when data are low-statistics. In all these cases, the covariance matrix would give symmetric error bars that do not accurately reflect the actual uncertainty.

10.7 When does profiling fail?

The $\Delta = 1$ rule relies on Wilks' theorem, which can fail when:

- the sample size is very small,
- parameters lie on or near boundaries,
- the likelihood is highly irregular.

In these cases, the profile likelihood is still useful as a diagnostic, but the mapping from Δ to confidence levels may require simulation (toy Monte Carlo studies).

11 Goodness of fit

After fitting, we need to assess whether the model actually describes the data. This section starts from the basics of χ^2 and builds toward a robust method for Poisson data.

11.1 Where the χ^2 distribution comes from

Suppose we have N independent measurements y_i, each Gaussian-distributed around a model prediction μ_i with known variance σ_i^2. Define the standardized residuals:

$$z_i = \frac{y_i - \mu_i}{\sigma_i}. \tag{33}$$

If the model is correct, each z_i is drawn from a standard normal $\mathcal{N}(0, 1)$. From the derivation of the likelihood estimate for gaussian noise previously, our loss function to minimize is

$$\chi^2 = \sum_{i=1}^{N} z_i^2 = \sum_{i=1}^{N} \frac{(y_i - \mu_i)^2}{\sigma_i^2}. \tag{34}$$

We can recognize that the z_i^2 term represents a distance, which is called the Mahalanobis distance. This is essentially the distance any particular value is from the mean of the distribution, see Fig. 3. The distribution of these distances is referred to as a χ^2 distribution with N degrees of freedom. Said more precisely, the χ^2 distribution is the distribution of a sum of squared standard normal variables. So the χ^2 test *statistic* above is drawn from an underlying χ^2 *distribution*.

If the model has p parameters that were estimated from the data, the effective degrees of freedom are reduced to $\nu = N - p$ (because the fitted parameters absorb some of the "disagreement" between data and model).

The chi-square distribution has well know properties, with an expectation value and variance given by the DOF and twice the DOF respectively:

$$\text{with } \mathbb{E}[\chi^2] = \nu, \quad \mathrm{Var}(\chi^2) = 2\nu. \tag{35}$$

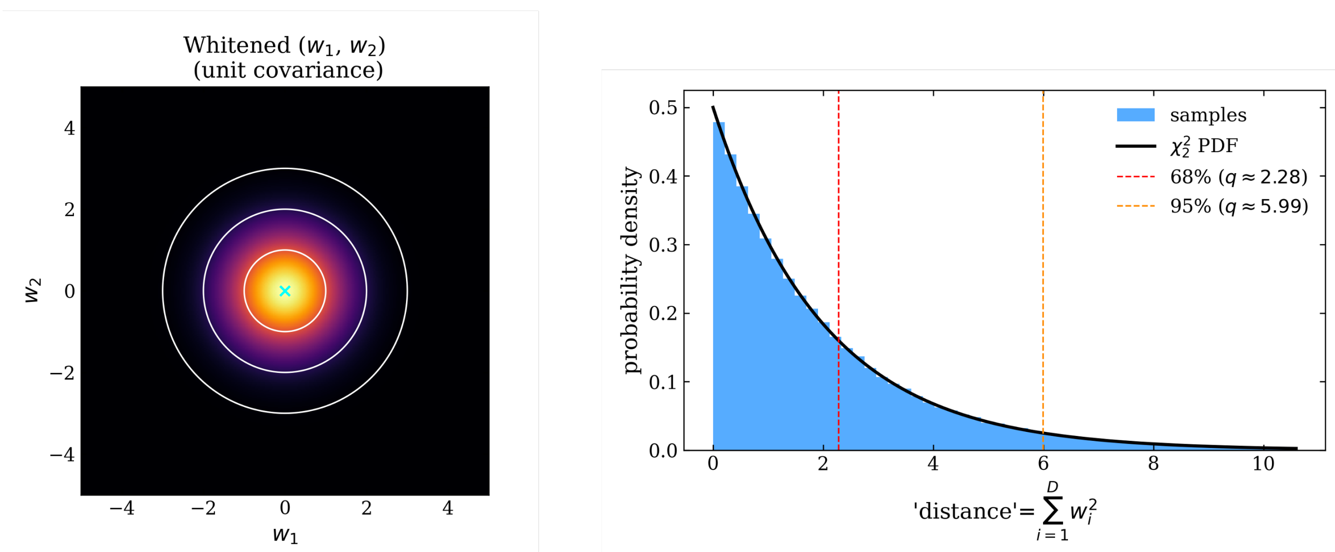

Figure 3: Contour lines for 68% and 95% level sets, and the distribution of distances from the mean, to illustrate the χ^2 distribution. Note, The x-axis uses w for distance and the text uses z... I just noticed this and don't want to remake my old figure...

A key point is that this result depends entirely on the data being Gaussian with known variances. **The χ^2 distribution is often used as a generic goodness-of-fit test, but this is only justified under Gaussian assumptions. For Poisson data, it can give misleading p-values.**

11.2 Quantifying the fit quality

This now gives us a robust method to check the 'goodness of fit' for a particular model. Our Given an observed χ^2 and ν degrees of freedom, we can compute a p-value:

$$p = P(\chi^2_\nu \geq \chi^2_{\text{obs}}). \tag{36}$$

This is the probability that a correct model would produce a χ^2 at least as large as what we observed. A very small p-value (say, $p < 0.01$) indicates that the model is a poor description of the data. A p-value near 1 indicates an unusually good fit, which may suggest that the uncertainties have been overestimated.

A more intuitive (and equivalent) way to think about this is to look at the chi-square distribution itself in Fig. 4. Since the χ^2 distribution is describing the sum of squares, the cumulative effect of this is that for a large number of degrees of freedom, this distribution approaches a Gaussian with $\mathcal{N}(\nu, 2\nu)$. For a normal distribution, we know that roughly 99.5% of all data should be within 3σ of the mean. Thus, as a rule of thumb, we can say that if our χ^2 test statistic from our fit is not within this range, our model is likely not a good description of the data.

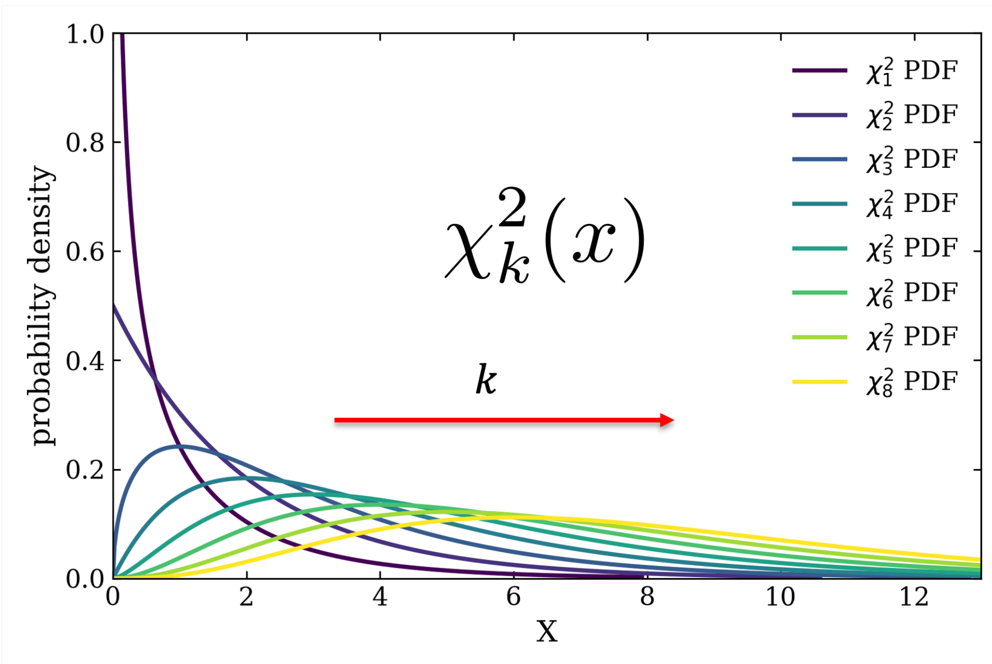

Figure 4: Chi2 PDF's for increasing degree's of freedom.

This is more informative than the commonly used "reduced χ^2 (χ^2/ν)", which discards the information about how wide the χ^2_ν distribution is. A reduced χ^2 of 1.3 could be perfectly acceptable for $\nu = 10$ but highly significant for $\nu = 1000$.

11.3 Why χ^2 fails for Poisson data

For Poisson-distributed counts, the Gaussian approximation breaks down when counts are small. Specifically:

- The variance $\text{Var}(n_i) = \mu_i$ depends on the model prediction, so it is not a known constant.
- The distribution of n_i is discrete and asymmetric for small μ_i.
- Bins with $n_i = 0$ or $n_i = 1$ do not produce z_i values that are remotely Gaussian.

As a result, $\sum(n_i - \mu_i)^2/\mu_i$ does not follow a χ^2 distribution when some bins have low counts. Using it as if it does will give unreliable p-values.

11.4 The deviance: a likelihood-based alternative

The Poisson analog of χ^2 is the **deviance**. It is defined as twice the log-likelihood ratio between the **saturated model** (a model with one free parameter per bin, i.e., a perfect fit where $\hat{\mu}_i = n_i$) and the fitted model:

$$D = 2 \sum_{i:\, n_i > 0} \left[n_i \log \frac{n_i}{\mu_i} - (n_i - \mu_i) \right]. \tag{37}$$

The deviance measures how much likelihood is lost by using the fitted model instead of the saturated model. If the fitted model is correct, each term in the sum is small on average, and D is small.

By Wilks' theorem, in the large-sample limit:

$$D \overset{\text{approx}}{\sim} \chi^2_\nu, \quad \nu = N_{\text{bins}} - N_{\text{params}}. \tag{38}$$

This is an asymptotic result which it holds when all μ_i are large. For data with many low-count bins, the approximation can be poor, and the p-value derived from the χ^2_ν distribution may not be reliable.

11.5 Parametric bootstrap for low data counts

When the asymptotic approximation is not trustworthy, we can build the reference distribution for D empirically. Straightforward: if we knew the true model, we could simulate many datasets from it and see how D is distributed. We do not know the true model, but our best-fit model is our best estimate of it.

The procedure is:

1. Fit the model to the real data, obtaining $\hat{\boldsymbol{\theta}}$ and $\mu_i(\hat{\boldsymbol{\theta}})$.
2. Compute the observed deviance D_{obs}.
3. For $k = 1, \ldots, N_{\text{sim}}$:
   (a) Generate a fake dataset: $n_i^{(k)} \sim \text{Poisson}(\mu_i(\hat{\boldsymbol{\theta}}))$.
   (b) Refit the model to the fake dataset, obtaining $\hat{\boldsymbol{\theta}}^{(k)}$ and $\mu_i^{(k)}$.
   (c) Compute the deviance $D^{(k)}$ between the fake data and the refitted model.
4. The p-value is the fraction of simulated deviances that exceed the observed deviance:

$$p = \frac{1}{N_{\text{sim}}} \sum_{k=1}^{N_{\text{sim}}} \mathbf{1}\left[D^{(k)} \geq D_{\text{obs}} \right]. \tag{39}$$

This method (while completely overkill for most things....) works regardless of the number of parameters or the shape of the likelihood. It is not hard to do, but comes with some computational cost, you must run the fit N_{sim} times (typically 500–1000).

The resulting distribution of $\{D^{(k)}\}$ can also be compared to the χ^2_ν distribution to check whether the asymptotic approximation is adequate for your particular dataset. If the two agree, you can use the analytic χ^2_ν p-value in the future with confidence. If they disagree, you know that the bootstrap p-value is the one to trust.

An example of this process is illustrated in Fig. 5 for the case of a model that is a poor description of the data and a model that is in agreement with the data. Also, notice that the data count is low enough, that the asymptotic approximation for the χ^2 is not valid.

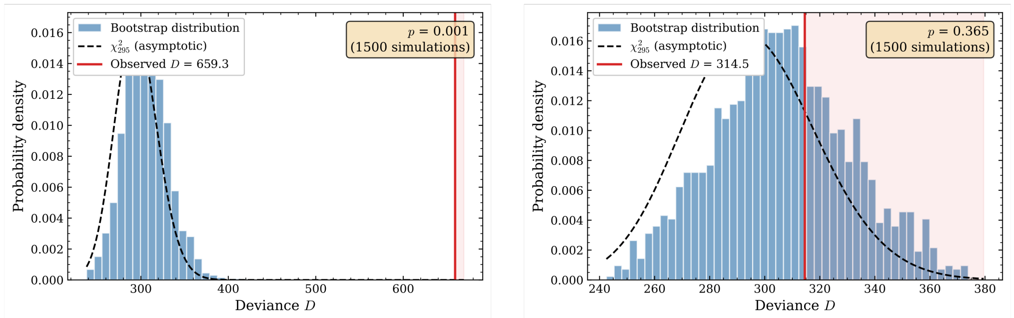

Figure 5: Parametric bootstrap example for a dataset that is not in good agreement with the model (left) and for a model with good agreement (right).

11.6 Inspecting residuals

The previous goodness-of-fit statistics are descriptions of how well the data and model globally agree, but offer very little in the way of diagnostics when there is disagreement. Therefor, visual inspection of the residuals is important. The standardized Poisson residuals,

$$r_i = \frac{n_i - \mu_i}{\sqrt{\mu_i}}, \tag{40}$$

should scatter around zero with roughly unit variance if the model is adequate. Systematic patterns, continuous sections of positive or negative residuals, residuals that grow toward the edges of the fit range, residuals that cluster near the peak, indicate that the model is missing something.

Examples of the residuals can be seen in Fig. 1. Notice how the least-squares (worse) fit has a lot of high variance points in the section of the peak itself, where as the Poisson residuals are all roughly within 2σ. Note that the visible bands in the Poisson residuals are actually an artifact of the low data bins, and is to be expected.

12 Summary and workflow

The recommended workflow for fitting count data is:

1. Specify a physically motivated model $\mu_i(\boldsymbol{\theta})$.
2. Use the Poisson likelihood (not χ^2) as the objective function.
3. Fit by minimizing $-\log \mathcal{L}$.
4. Inspect residuals and compute the deviance to assess fit quality.
5. Estimate uncertainties using the covariance matrix for a quick estimate, or the profile likelihood for robust results.

Key takeaways:

- The Poisson likelihood is the correct objective function for count data. Least-squares methods are approximations that work well at high counts but can fail at low counts.

- The covariance matrix gives fast, symmetric error bars based on a local quadratic approximation. It is usually adequate when statistics are high.
- The profile likelihood gives robust, potentially asymmetric error bars that account for parameter correlations and nonlinearities. It is the right tool when statistics are low or when parameters are near boundaries.
- The fit function $\mu_i(\boldsymbol{\theta})$ is not just a curve, it is the expected value entering the statistical model. Getting the fit function right and getting the statistical model right are equally important.

  A parameter estimate is only meaningful if both the physical model and the statistical model reflect the data-generating process.

13 Further reading

- **F. James, *Statistical Methods in Experimental Physics*, 2nd ed. (World Scientific, 2006).** The standard reference for statistical methods in physics, written by the author of the MINUIT minimization package. Chapters 9–10 on maximum likelihood and least squares are directly relevant to this note. The treatment of goodness of fit and the comparison of different fitting methods is more thorough than what most textbooks offer. Highly recommended as a desk reference, but also written at a fairly fundamental/advanced level.
- **G. Cowan, *Statistical Data Analysis* (Oxford University Press, 1998).** A concise and well-written introduction aimed at physical scientists. Covers parameter estimation, hypothesis testing, confidence intervals, and Monte Carlo methods. Chapters 6–7 on maximum likelihood and hypothesis testing give a clear, compact treatment of much of what this note covers. More accessible than James for a first read.
- **Particle Data Group, "Statistics" review, Chapter 40 in the *Review of Particle Physics*.** A comprehensive survey of statistical methods used in high-energy physics, updated regularly. Covers frequentist and Bayesian inference, the profile likelihood ratio, the look-elsewhere effect, and the treatment of nuisance parameters. Freely available at `https://pdg.lbl.gov/2024/reviews/rpp2024-rev-statistics.pdf`. This is the reference that working experimentalists in particle physics typically use in practice.
- **G. Cowan, K. Cranmer, E. Gross, and O. Vitells, "Asymptotic formulae for likelihood-based tests of new physics," *Eur. Phys. J. C* 71, 1554 (2011).** The paper that formalized the profile likelihood ratio test statistic and its asymptotic distributions, including the treatment of the look-elsewhere effect for continuous search parameters. This is the technical basis for the discovery statistics used in the Higgs boson search and many other analyses.
- **S. Baker and R. D. Cousins, "Clarification of the use of CHI-square and likelihood functions in fits to histograms," *Nucl. Instrum. Methods* 221, 437 (1984).** A short, influential paper that clearly explains why Poisson likelihood fitting is preferable to χ^2 fitting for binned count data.